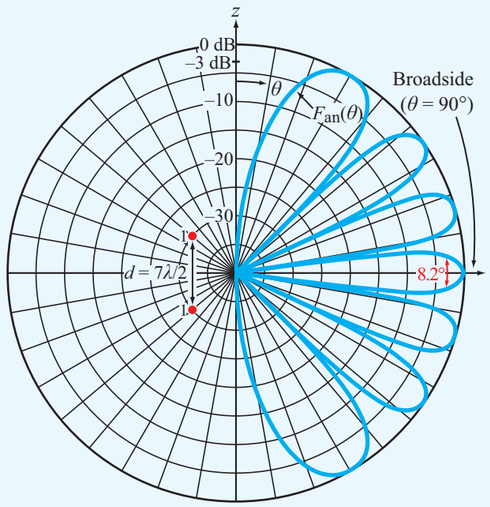


Figure 9-31 Normalized array pattern of a two-element array with spacing $d = 7\lambda/2$.

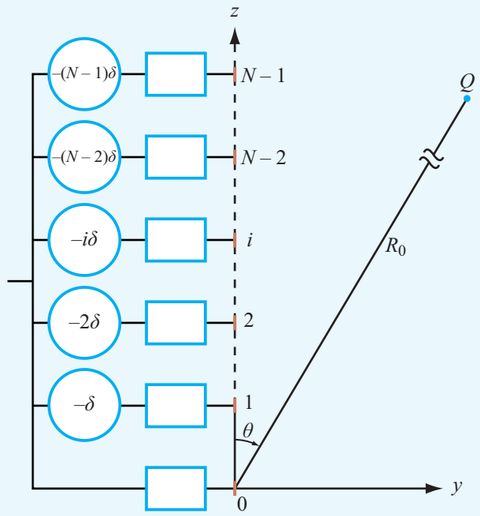


Figure 9-32 The application of linear phase.

separation between the array elements, d, measured in units of $\lambda/2$.

9-11 Electronic Scanning of Arrays

The discussion in the preceding section was concerned with uniform-phase arrays, in which the phases of the feeding coefficients, ψ_0 to ψ_{N-1}, are all equal. In this section, we examine the use of phase delay between adjacent elements as a tool to ***electronically steer*** the direction of the array-antenna beam from broadside at $\theta = 90°$ to any desired angle θ_0. In addition to eliminating the need to mechanically steer an antenna to change its beam's direction, electronic steering allows beam scanning at very fast rates.

► Electronic steering is achieved by applying a ***linear phase distribution*** across the array: $\psi_0 = 0$, $\psi_1 = -\delta$, $\psi_2 = -2\delta$, etc. ◄

As shown in **Fig. 9-32**, the phase of the ith element relative to that of the zeroth element is

$$\psi_i = -i\delta, \tag{9.116}$$

where δ is the ***incremental phase delay*** between adjacent elements. Use of Eq. (9.116) in Eq. (9.104) leads to

$$F_a(\theta) = \left|\sum_{i=0}^{N-1} a_i e^{-ji\delta} e^{jikd\cos\theta}\right|^2 = \left|\sum_{i=0}^{N-1} a_i e^{ji(kd\cos\theta-\delta)}\right|^2 = \left|\sum_{i=0}^{N-1} a_i e^{ji\gamma'}\right|^2 = F_a(\gamma'), \tag{9.117}$$

where we introduced a new variable given by

$$\gamma' = kd\cos\theta - \delta. \tag{9.118}$$

For reasons that become clear later, we define the phase shift δ in terms of an angle θ_0, which we call the ***scan angle***, as follows:

$$\delta = kd\cos\theta_0. \tag{9.119}$$

Hence, γ' becomes

$$\gamma' = kd(\cos\theta - \cos\theta_0). \tag{9.120}$$

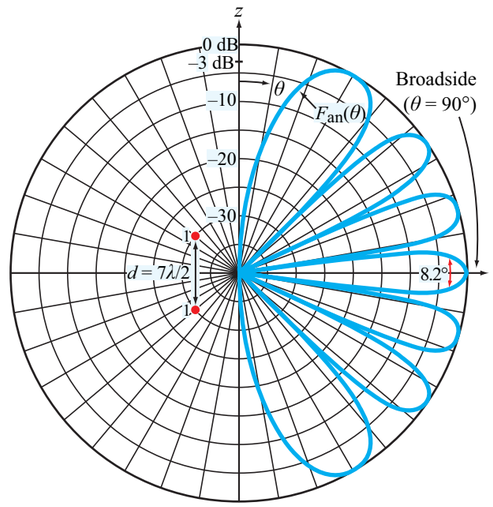


图 9-31 间距为 d = 7λ/2 的二元素数组的归一化数组模式。

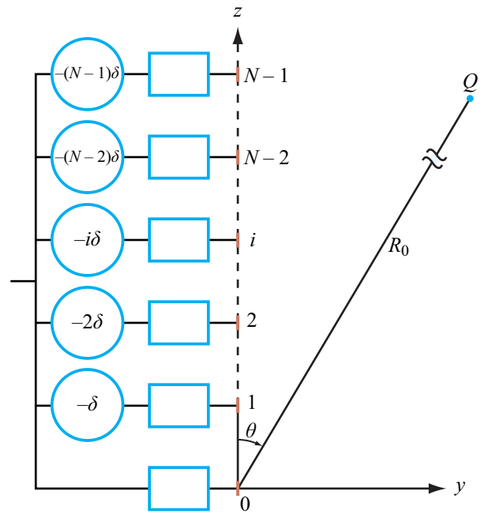


图9-32 线性相位的应用。

数组元素之间的间隔 d，以 λ/2 为单位测量。

9-11 阵列的电子扫描

上一节中的讨论涉及均匀相位阵列，其中馈电系数 ψ_0 至 ψ_{N-1} 的相位均相等。在本节中，我们将研究如何使用相邻元件之间的相位延迟作为一种工具，以电子方式将阵列天线波束的方向从 θ = 90° 的侧面转向任何所需的角度 θ_0。除了无需机械操纵天线来改变波束方向外，电子操纵还可以以非常快的速率进行波束扫描。

► 电子转向是通过在阵列上应用线性相位分布来实现的：$\psi_0 = 0$、$\psi_1 = -\delta$、$\psi_2 = -2\delta$ 等。◄

如图9-32所示，第i个单元相对于第0个单元的相位为

$$\psi_i = -i\delta, \tag{9.116}$$

其中 δ 是相邻元件之间的增量相位延迟。使用等式。 (9.116) 式(9.104) 导致

$$F_a(\theta) = \left|\sum_{i=0}^{N-1} a_i e^{-ji\delta} e^{jikd\cos\theta}\right|^2 = \left|\sum_{i=0}^{N-1} a_i e^{ji(kd\cos\theta-\delta)}\right|^2 = \left|\sum_{i=0}^{N-1} a_i e^{ji\gamma'}\right|^2 = F_a(\gamma'), \tag{9.117}$$

我们引入了一个新变量，由下式给出

$$\gamma' = kd\cos\theta - \delta. \tag{9.118}$$

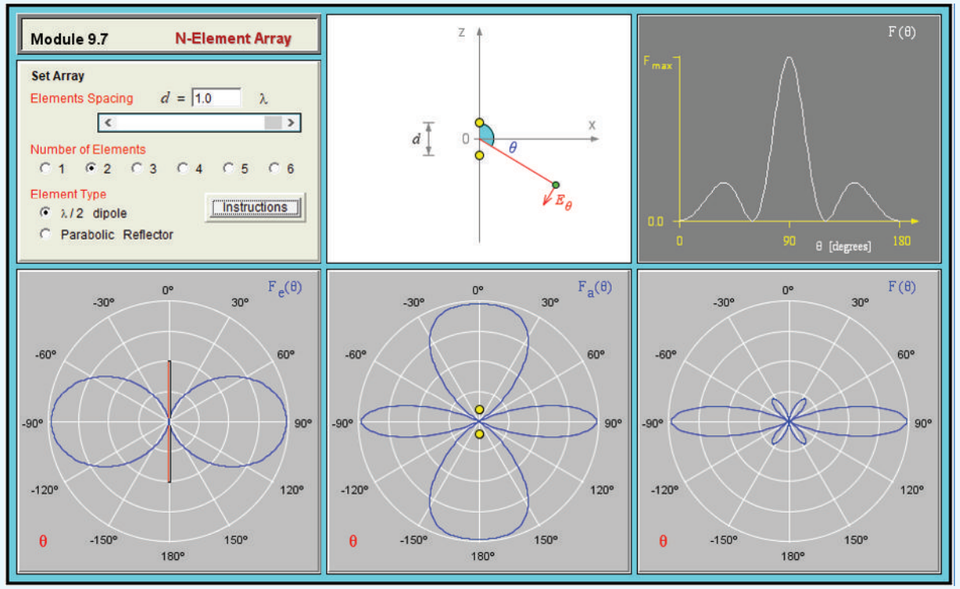
出于稍后会清楚的原因，我们用角度 θ_0 定义相移 δ，我们将其称为扫描角，如下所示：

$$\delta = kd\cos\theta_0. \tag{9.119}$$

因此，γ' 变为

$$\gamma' = kd(\cos\theta - \cos\theta_0). \tag{9.120}$$

Module 9.7 ***N*-Element Array** This module displays the far-field patterns of an array of N identical, equally spaced antennas, with N being a selectable integer between 1 and 6. Two types of antennas can be simulated: $\lambda/2$-dipoles and parabolic reflectors. This module provides visual examples of the pattern multiplication principle.



The array factor given by Eq. (9.117) has the same functional form as the array factor developed earlier for the uniform-phase array [see Eq. (9.107)], except that γ is replaced with γ'. Hence,

► Regardless of the amplitude distribution across an array, its array factor $F_a(\gamma')$, when excited by a linear-phase distribution, can be obtained from $F_a(\gamma)$, the expression developed for the array assuming a uniform-phase distribution, by replacing γ with γ'. ◄

If the amplitude distribution is symmetrical with respect to the array center, the array factor $F_a(\gamma')$ is maximum when its argument $\gamma' = 0$. When the phase is uniform ($\delta = 0$), this condition corresponds to the direction $\theta = 90°$, which is why the uniform-phase arrangement is called a broadside array. According to Eq. (9.120), in a linearly phased array, $\gamma' = 0$ when $\theta = \theta_0$. Thus, by applying linear phase across the array, the array pattern is shifted along the $\cos\theta$ axis by an amount $\cos\theta_0$, and the direction of maximum radiation is ***steered*** from the broadside direction ($\theta = 90°$) to the direction $\theta = \theta_0$. To steer the beam all the way to the ***end-fire direction*** ($\theta = 0$), the incremental phase shift δ should be equal to kd radians.

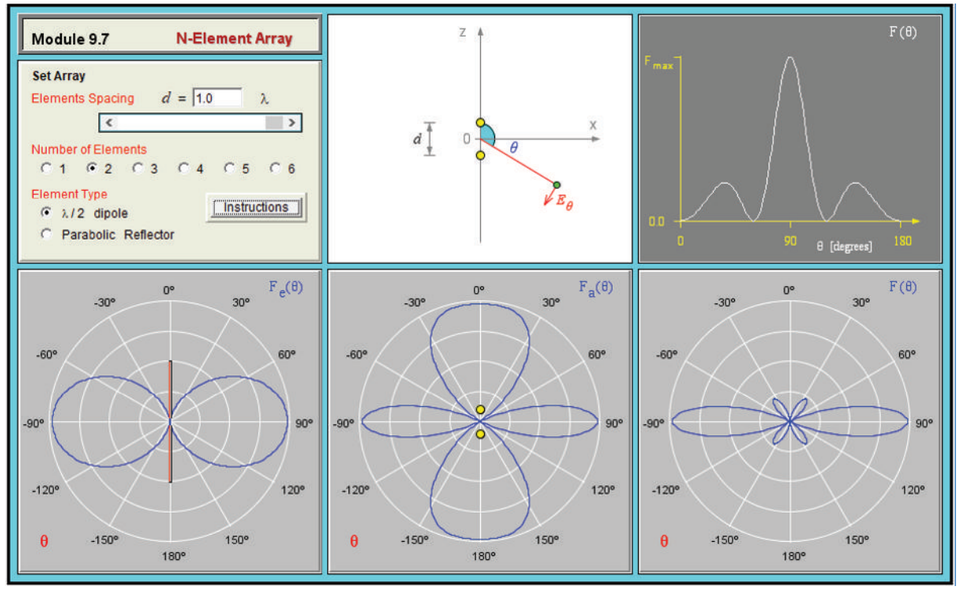
9-11.1 Uniform-Amplitude Excitation

To illustrate the process with an example, consider the case of the N-element array excited by a uniform-amplitude distribution. Its normalized array factor is given by Eq. (9.115). Upon replacing γ with γ', we have

$$F_{an}(\gamma') = \frac{\sin^2(N\gamma'/2)}{N^2\sin^2(\gamma'/2)} \qquad (9.121)$$

with γ' as defined by Eq. (9.120). For an array with $N = 10$ and $d = \lambda/2$, plots of the main lobe of $F_{an}(\theta)$ are shown in **Fig. 9-33** for $\theta_0 = 0°$, 45°, and 90°. We note that the half-power beamwidth increases as the array beam is steered from broadside to end-fire directions.

模块 9.7 N 元阵列 该模块显示 N 个相同、等距天线阵列的远场方向图，N 是 1 到 6 之间的可选整数。可以模拟两种类型的天线：λ/2 偶极子和抛物面反射器。该模块提供了模式乘法原理的直观示例。



方程给出的阵列因子。（9.117）与之前为均匀相位阵列开发的阵列因子具有相同的函数形式[见方程1]。（9.107)]，只不过 γ 被替换为 γ′。因此，

►无论阵列上的幅度分布如何，其阵列因子 $F_a(\gamma')$，当由线性相位分布激励时，可以从 F 获得$_a(\gamma)$，为假设均匀相位分布的阵列开发的表达式，通过替换γ和 γ′. ◄

如果振幅分布相对于阵列中心对称，则当其自变量γ′ =为0时，阵列因子$F_a(\gamma')$最大。当相位均匀（δ = 0）时，该条件对应于方向θ =90°，这就是为什么均匀相位排列被称为宽边阵列的原因。根据方程。（9.120），在线性相控阵中，当 θ = θ_0 时，γ′ = 0。因此，通过在阵列上应用线性相位，阵列图案沿 cos θ 轴移动一定量

cos θ_0，最大辐射方向从舷侧方向（θ = 90°）转向θ = θ_0 方向。为了将光束一直转向端射方向 (θ = 0)，增量相移 δ 应等于 kd 弧度。

9-11.1 等幅激励

为了用一个例子来说明这个过程，考虑由均匀幅度分布激励的 N 元阵列的情况。其归一化阵列因子由等式给出。（9.115）。将 γ 替换为 γ′ 后，我们有

$$F_{an}(\gamma') = \frac{\sin^2(N\gamma'/2)}{N^2\sin^2(\gamma'/2)} \qquad (9.121)$$

与 γ′ 如方程所定义。（9.120）。对于 N = 10 和 d = λ/2 的阵列，对于 $\theta_0 = 0°$、45° 和 90° 的 $F_{an}(\theta)$ 主瓣图如图 9-33 所示。我们注意到，随着阵列波束从侧面转向端射方向，半功率波束宽度增加。

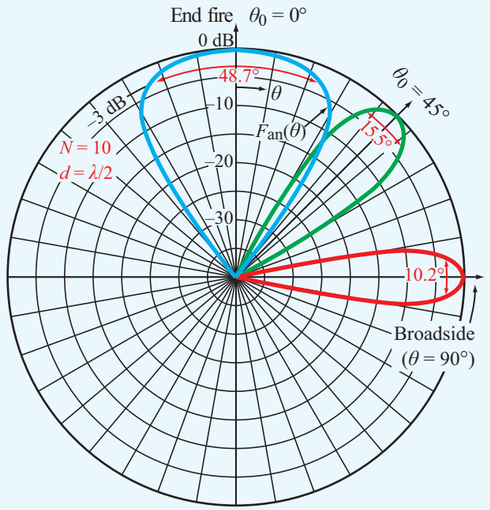


Figure 9-33 Normalized array pattern of a 10-element array with $\lambda/2$ spacing between adjacent elements. All elements are excited with equal amplitude. Through the application of linear phase across the array, the main beam can be steered from the broadside direction ($\theta_0 = 90°$) to any scan angle θ_0. Equiphase excitation corresponds to $\theta_0 = 90°$.

9-11.2 Array Feeding

According to the foregoing discussion, to steer the antenna beam to an angle θ_0, two conditions must be met: (1) the phase distribution must be linear across the array and (2) the magnitude of the incremental phase delay δ must satisfy Eq. (9.119). The combination of these two conditions provides the necessary tilting of the beam from $\theta = 90°$ (broadside) to $\theta = \theta_0$. This can be accomplished by controlling the excitation of each radiating element individually through the use of electronically controlled phase shifters. Alternatively, a technique known as ***frequency scanning*** can be used to provide control of the phases of all the elements simultaneously. **Figure 9-34** shows an example of a simple feeding arrangement employed in frequency-scanning arrays. A common feed point is connected to the radiating elements through transmission lines of varying lengths. Relative to the zeroth element, the path between the common feed point and a radiating element is longer by l for the first element, by $2l$ for the second, and by $3l$ for the third.

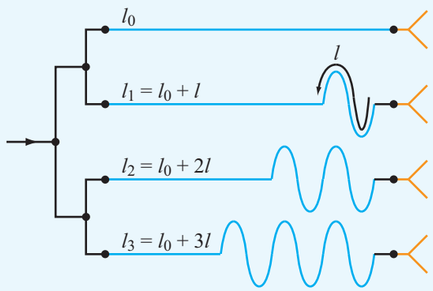


Figure 9-34 An example of a feeding arrangement for frequency-scanned arrays.

Thus, the path length for the ith element is

$$l_i = il + l_0, \tag{9.122}$$

where l_0 is the path length of the zeroth element. Waves of frequency f propagating through a transmission line of length l_i are characterized by a phase factor $e^{-j\beta l_i}$, where $\beta = 2\pi f/u_p$ is the phase constant of the line and u_p is its phase velocity. Hence, the *incremental* phase delay of the ith element relative to the phase of the zeroth element is

$$\psi_i(f) = -\beta(l_i - l_0) = -\frac{2\pi}{u_p} f(l_i - l_0) = -\frac{2\pi i}{u_p} fl. \tag{9.123}$$

Suppose that at a given reference frequency f_0 we choose the incremental length l so that

$$l = \frac{n_0 u_p}{f_0}, \tag{9.124}$$

where n_0 is a *specific* positive integer. In this case, the phase delay $\psi_1(f_0)$ becomes

$$\psi_1(f_0) = -2\pi\left(\frac{f_0 l}{u_p}\right). = -2n_0\pi \tag{9.125}$$

Similarly, $\psi_2(f_0) = -4n_0\pi$ and $\psi_3(f_0) = -6n_0\pi$. That is, at f_0 all the elements have equal phase (within multiples of 2π) and the array radiates in the broadside direction. If f is changed to

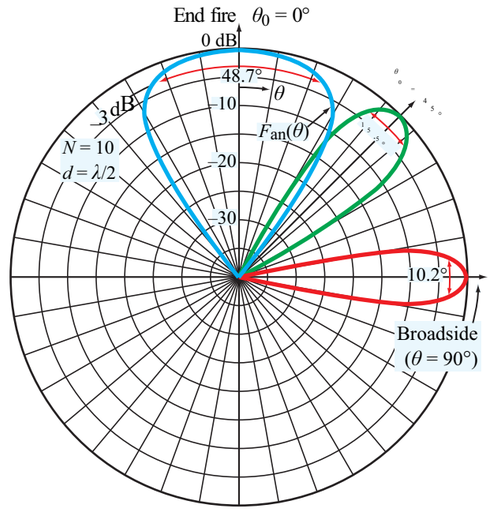


图 9-33 10 元素数组的标准化数组模式，相邻元素之间的间距为 λ/2。所有元素都以相等的幅度被激发。通过在阵列上应用线性相位，主波束可以从侧边方向 ($\theta_0 = 90°$) 转向任何扫描角度 θ_0。等相激励对应于 $\theta_0 = 90°$。

9-11.2 数组馈送

根据前面的讨论，为了将天线波束转向θ_0角度，必须满足两个条件：（1）整个阵列的相位分布必须是线性的，（2）增量相位延迟δ的幅度必须满足方程1。（9.119）。这两个条件的组合提供了光束从θ = 90° (侧边)到θ = θ_0 的必要倾斜。这可以通过使用电子控制移相器单独控制每个辐射元件的激励来实现。或者，可以使用称为频率扫描的技术来同时提供对所有元件的相位的控制。图 9-34 显示了频率扫描阵列中采用的简单馈电布置的示例。公共馈电点通过不同长度的传输线连接到辐射元件。相对于第 0 个元件，公共馈电点和辐射元件之间的路径第一个元件长 l，第二个元件长 2l，第三个元件长 3l。

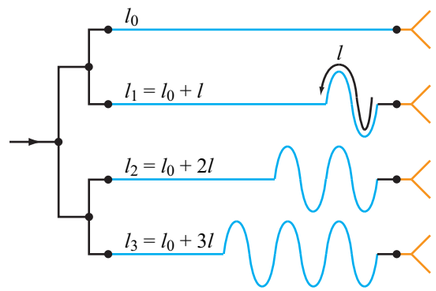


图 9-34 频率扫描阵列的馈电布置示例。

因此，第 i 个元素的路径长度为

$$l_i = il + l_0, \tag{9.122}$$

其中 l_0 是第零个元素的路径长度。通过长度为 l_i 的传输线传播的频率为 f 的波由相位因子 $e^{-j\beta l_i}$ 表征，其中 $\beta = 2\pi f/u_p$ 是传输线的相位常数，u_p 是其相速度。因此，第 i 个元件相对于第 0 个元件的相位延迟增量为

$$\psi_i(f) = -\beta(l_i - l_0) = -\frac{2\pi}{u_p} f(l_i - l_0) = -\frac{2\pi i}{u_p} fl. \tag{9.123}$$

假设在给定的参考频率 f_0 我们选择增量长度 l 使得

$$l = \frac{n_0 u_p}{f_0}, \tag{9.124}$$

其中 n_0 是一个特定的正整数。在这种情况下，相位延迟 $\psi_1(f_0)$ 变为

$$\psi_1(f_0) = -2\pi\left(\frac{f_0 l}{u_p}\right). = -2n_0\pi \tag{9.125}$$

类似地，$\psi_2(f_0) = -4n_0\pi$ 和 $\psi_3(f_0) = -6n_0\pi$。也就是说，在 f_0 处，所有单元都具有相等的相位（在 2π 的倍数内），并且阵列沿宽边方向辐射。如果 f 改为

$f_0 + \Delta f$, the new phase shift of the first element relative to the zeroth element is

$$\begin{aligned}\psi_1(f_0+\Delta f) &= -\frac{2\pi}{u_p}(f_0+\Delta f)l \\ &= -\frac{2\pi f_0 l}{u_p} - \left(\frac{2\pi l}{u_p}\right)\Delta f \\ &= -2n_0\pi - 2n_0\pi\left(\frac{\Delta f}{f_0}\right) \\ &= -2n_0\pi - \delta, \end{aligned} \tag{9.126}$$

where use was made of Eq. (9.124) and δ is defined as

$$\delta = 2n_0\pi\left(\frac{\Delta f}{f_0}\right). \tag{9.127}$$

Similarly, $\psi_2(f_0+\Delta f) = 2\psi_1$ and $\psi_3(f_0+\Delta f) = 3\psi_1$. Ignoring the factor of 2π and its multiples (since they exercise no influence on the relative phases of the radiated fields), we see that the incremental phase shifts are directly proportional to the fractional frequency deviation $(\Delta f/f_0)$. Thus, in an array with N elements, controlling Δf provides a direct control of δ, which in turn controls the scan angle θ_0 according to Eq. (9.119). Equating Eq. (9.119) to Eq. (9.127) and then solving for $\cos\theta_0$ leads to

$$\cos\theta_0 = \frac{2n_0\pi}{kd}\left(\frac{\Delta f}{f_0}\right) \tag{9.128}$$

As f is changed from f_0 to $f_0+\Delta f$, $k = 2\pi/\lambda = 2\pi f/c$ also changes with frequency. However, if $\Delta f/f_0$ is small, we may treat k as a constant equal to $2\pi f_0/c$; the error in $\cos\theta_0$ resulting from the use of this approximation in Eq. (9.128) is on the order of $\Delta f/f_0$.

Example 9-8: Electronic Steering

Design a steerable six-element array with the following specifications:

1. All elements are excited with equal amplitudes.
2. At $f_0 = 10$ GHz, the array radiates in the broadside direction, and the interelement spacing $d = \lambda_0/2$, where $\lambda_0 = c/f_0 = 3$ cm.
3. The array pattern is to be electronically steerable in the elevation plane over the angular range extending between $\theta_0 = 30°$ and $\theta_0 = 150°$.
4. The antenna array is fed by a voltage-controlled oscillator whose frequency can be varied over the range from 9.5 to 10.5 GHz.
5. The array uses a feeding arrangement of the type shown in Fig. 9-34, and the transmission lines have a phase velocity $u_p = 0.8c$.

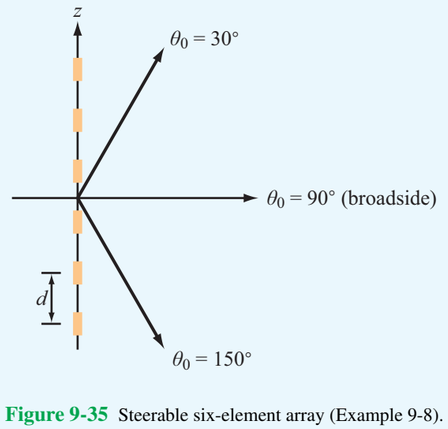


Figure 9-35 Steerable six-element array (Example 9-8).

Solution: The array is to be steerable from $\theta_0 = 30°$ to $\theta_0 = 150°$ (Fig. 9-35). For $\theta_0 = 30°$ and

$$kd = \left(\frac{2\pi}{\lambda_0}\right)\left(\frac{\lambda_0}{2}\right) = \pi,$$

Eq. (9.128) gives

$$0.87 = 2n_0\left(\frac{\Delta f}{f_0}\right). \tag{9.129}$$

We are given that $f_0 = 10$ GHz and the oscillator frequency can be varied between $(f_0 - 0.5$ GHz) and $(f_0 + 0.5$ GHz). Thus, $\Delta f_{max} = 0.5$ GHz. To satisfy Eq. (9.129), we need to choose n_0 so that Δf is as close as possible to, but not larger than, Δf_{max}. Solving Eq. (9.129) for n_0 with $\Delta f = \Delta f_{max}$ gives

$$n_0 = \frac{0.87}{2}\frac{f_0}{\Delta f_{max}} = 8.7.$$

Since n_0 is not an integer, we need to modify its value by rounding it upward to the next whole-integer value. Hence, we set $n_0 = 9$.

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f0 + Δ f，第一个元素相对于第零个元素的新相移为

$$\begin{aligned}\psi_1(f_0+\Delta f) &= -\frac{2\pi}{u_p}(f_0+\Delta f)l \\ &= -\frac{2\pi f_0 l}{u_p} - \left(\frac{2\pi l}{u_p}\right)\Delta f \\ &= -2n_0\pi - 2n_0\pi\left(\frac{\Delta f}{f_0}\right) \\ &= -2n_0\pi - \delta, \end{aligned} \tag{9.126}$$

其中使用了方程式。 (9.124) 和 δ 定义为

$$\delta = 2n_0\pi\left(\frac{\Delta f}{f_0}\right). \tag{9.127}$$

相似地，$\psi_2(f_0+\Delta f) = 2\psi_1$和$\psi_3(f_0+\Delta f) = 3\psi_1$。忽略因子 2π 及其倍数（因为它们对辐射场的相对相位没有影响），我们看到增量相移与分数频率偏差成正比（$\Delta f/f_0$）。因此，在具有 N 个元素的数组中，控制Δf 提供直接控制 δ，进而控制扫描角度θ_0根据方程式（9.119）。等式(9.119) 至方程式(9.127) 然后求解 $\cos\theta_0$导致

$$\cos\theta_0 = \frac{2n_0\pi}{kd}\left(\frac{\Delta f}{f_0}\right) \tag{9.128}$$

当 f 从 f_0 变为 $f_0 + \Delta f$ 时，$k = 2\pi/\lambda = 2\pi f/c$ 也随频率变化。然而，如果$\Delta f / f_0$很小，我们可以将k视为等于$2\pi f_0/c$的常数； $\cos\theta_0$ 中的误差是由于在方程式中使用此近似值而产生的。 (9.128) 的量级为 $\Delta f / f_0$。

例 9-8：电子转向

设计一个具有以下规格的可操纵六元阵列：

1. 所有元件均以相等的幅度激励。
2. 在 $f_0 = 10$ GHz 处，阵列沿侧面方向辐射，元件间间距 $d = \lambda_0/2$，其中 $\lambda_0 = c/f_0 = 3$ cm。
3. 阵列方向图应在$\theta_0 = 30°$ 和$\theta_0 = 150°$ 之间延伸的角度范围内，在仰角平面中可电子操纵。
4. 天线阵列由压控振荡器供电，其频率可在 9.5 至 10.5 GHz 范围内变化。
5. 该阵列使用图9-34所示类型的馈电布置，并且传输线具有相速度$u_p = 0.8c$。

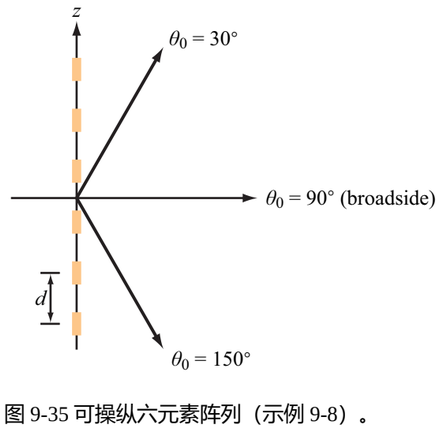


图 9-35 可操纵六元素阵列（示例 9-8）。

解决方案：阵列应可从 $\theta_0 = 30°$ 转向 $\theta_0 = 150°$ (。 9-35) 。对于 $\theta_0 = 30°$ 和

$$kd = \left(\frac{2\pi}{\lambda_0}\right)\left(\frac{\lambda_0}{2}\right) = \pi,$$

等式。 (9.128) 给出

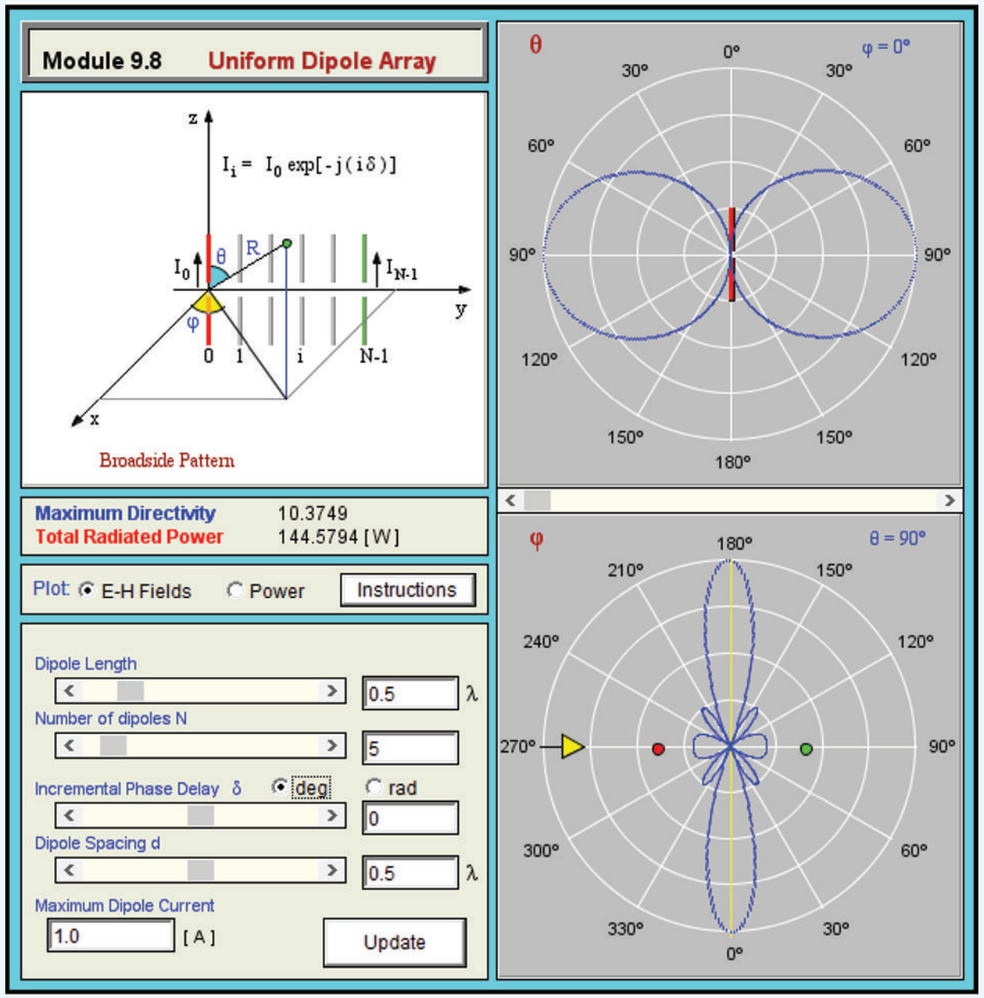
$$0.87 = 2n_0\left(\frac{\Delta f}{f_0}\right). \tag{9.129}$$

我们已知 $f_0 = 10$ GHz，并且振荡器频率可以在 ($f_0 - 0.5$ GHz) 和 ($f_0 + 0.5$ GHz) 之间变化。因此，$\Delta f_{max} = 0.5$ GHz。满足方程。 (9.129)，我们需要选择n_0，以便Δ f 尽可能接近但不大于Δf_{max}。求解方程。 (9.129) 对于 n_0 和 $\Delta f = \Delta f_{max}$ 给出

$$n_0 = \frac{0.87}{2}\frac{f_0}{\Delta f_{max}} = 8.7.$$

由于 n_0 不是整数，我们需要通过向上舍入到下一个整数值来修改它的值。因此，我们将 n_0 = 设置为 9。

Module 9.8 Uniform Dipole Array For an array of up to 50 identical vertical dipoles of selectable length and current maximum that is excited with incremental phase delay δ between adjacent elements, the module displays the elevation and azimuthal patterns of the array. By varying δ, the array pattern can be steered in the horizontal plane.



模块 9.8 均匀偶极子阵列 对于最多 50 个具有可选长度和电流最大值的相同垂直偶极子阵列，该阵列由相邻元件之间的增量相位延迟 δ 激励，该模块显示阵列的仰角和方位角图案。通过改变δ，可以在水平面上操纵阵列模式。

